Learning Outcome

- LO3 – implement basic bioinformatics analysis in scope of DNA composition and sequence analysis, protein synthesis, sequence alignment, and other works related to biological database using Biopython
- LO4 – analyze the basic bioinformatics analysis results using Biopython

Topic

- Session 03 – Protein Synthesis

Sub Topics

- Protein Synthesis
- Amino Acid Codon Table
- 3D Structure of Protein

Soal
Case

For given **DNA sequences**:

Sequence A:
ATGCGTTTATGACGTATCTGAAGTCTACTAGGGTGATCTGACTAA

Sequence B:
GTGTTAGTGCGCTAGTGCTGACTAGGTACAGTAGCTAGTTACTAG

And following **mRNA codon arrangements table**:

First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU, UUC Phe UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr **UAA Stop** **UAG Stop**	UGU, UGC Cys **UGA Stop** UGG Trp	U C A G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U C A G
A	AUU, AUC, AUA Ile **AUG Met**	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn AAA, AAG Lys	AGU, AGC Ser AGA, AGG Arg	U C A G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U C A G

Please use **Biopython** to:

- **Transcribe** the **DNA sequence A** and **sequence B** into **mRNA**.
- **Combine 24 starting** nucleotides of **mRNA sequence A** and **21 ending** nucleotides of **mRNA sequence B** and **set** it as **mRNA sequence C**.
- **Translate** the **mRNA sequence A**, **sequence B**, and **sequence C** into **amino acids sequences**.
- **Transcribe** back the **mRNA sequence C** into **DNA**.
- **Calculate** the **complementary sequences** of **DNA sequence A**, **sequence B**, and **sequence C**.